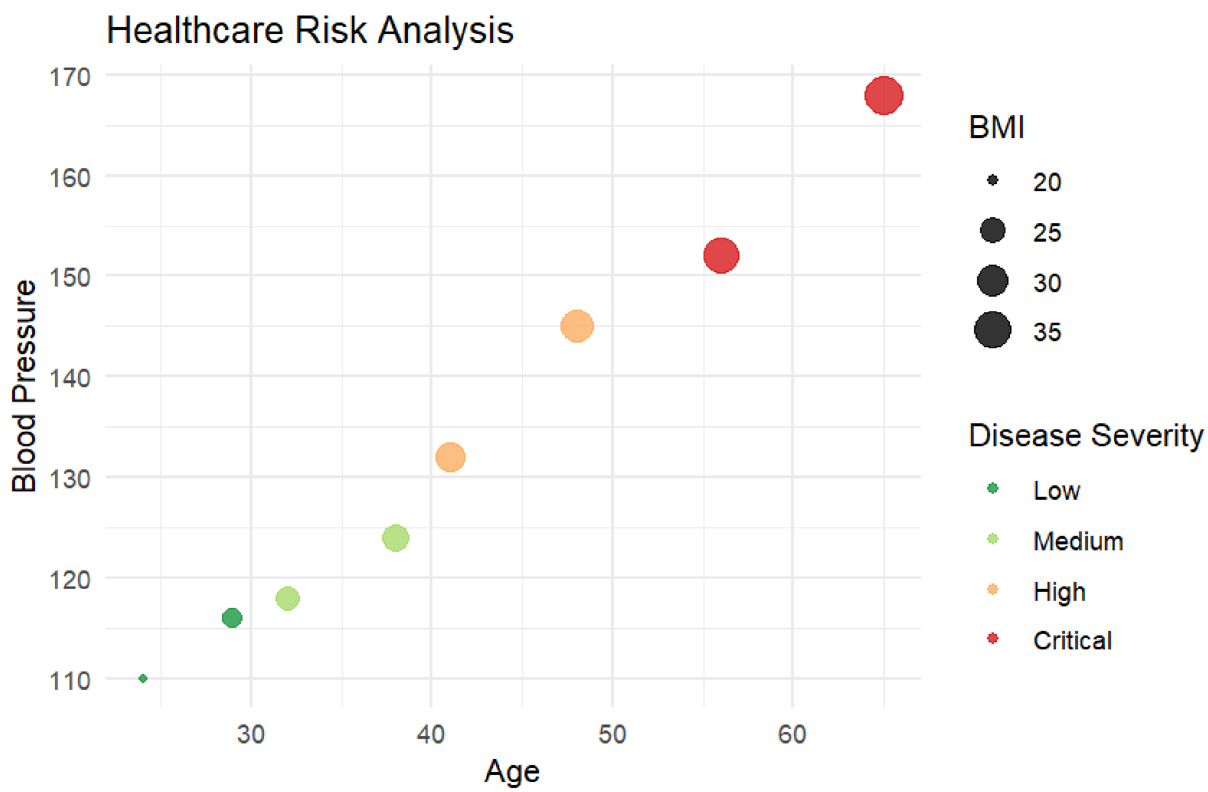
CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Question 1 – Healthcare Analytics



CODE:

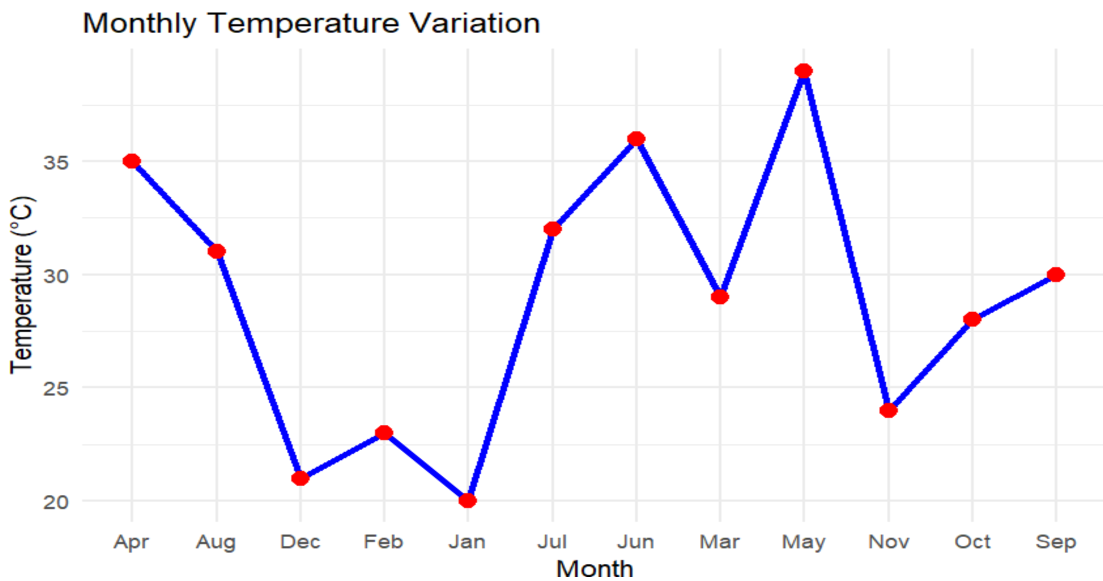
```
# Install package if needed
install.packages("ggplot2")

# Load library
library(ggplot2)

# Create dataset
health <- data.frame(
  Patient=c("P1","P2","P3","P4","P5","P6","P7","P8"),
  Age=c(24,32,41,48,56,65,38,29),
  BMI=c(19.8,24.5,28.1,30.3,33.2,36.5,26.2,22.1),
```

```r
  BloodPressure=c(110,118,132,145,152,168,124,116),
  Severity=factor(c("Low","Medium","High","High","Critical","Critical","Medium","Low"),
                  levels=c("Low","Medium","High","Critical"))
)
# Scatter Plot
ggplot(health,
       aes(x=Age,
           y=BloodPressure,
           color=Severity,
           size=BMI)) +
  geom_point(alpha=0.8) +
  scale_color_brewer(palette="RdYlGn", direction=-1) +
  labs(title="Healthcare Risk Analysis",
       x="Age",
       y="Blood Pressure",
       color="Disease Severity",
       size="BMI") +
  theme_minimal()
```

Justification

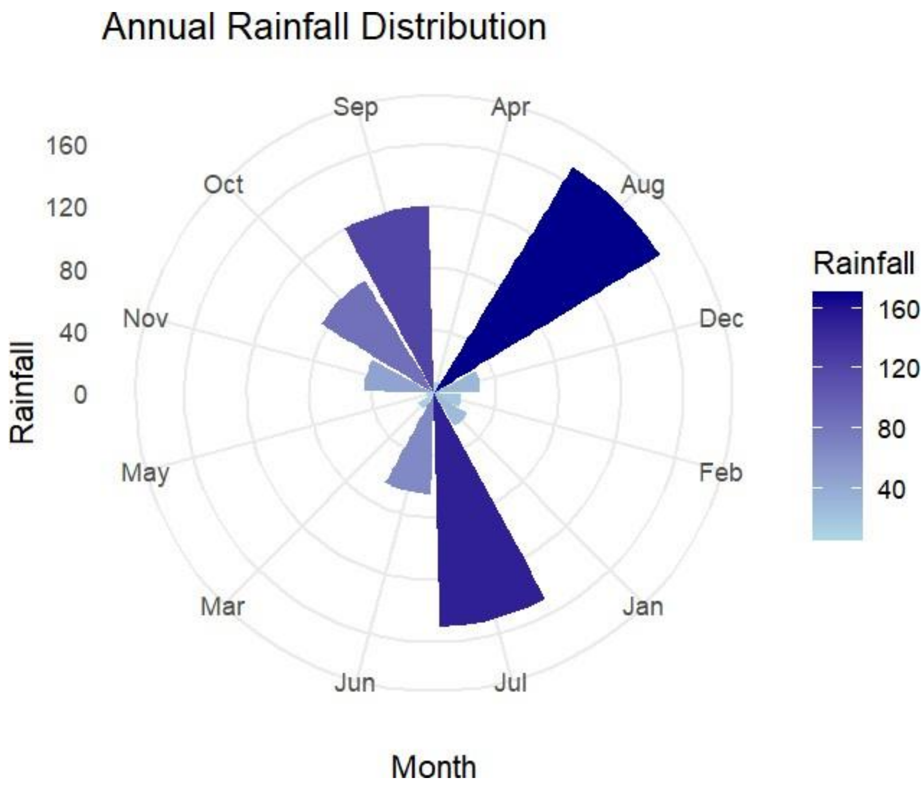
- Age → X-axis: Shows how health risk changes with age.
- Blood Pressure → Y-axis: Indicates cardiovascular health.
- Disease Severity → Color: Green to red clearly represents increasing risk.
- BMI → Point Size: Larger points indicate higher BMI and greater obesity risk.
- The scatter plot effectively shows relationships among four medical variables simultaneously.

Question 2 – Climate Science



CODE:

```
library(ggplot2)

climate <- data.frame(
  Month=c("Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec"),
  Solar=c(180,200,240,280,320,300,260,250,240,220,190,170),
  Temperature=c(20,23,29,35,39,36,32,31,30,28,24,21),
  Rainfall=c(25,18,12,8,5,65,150,170,120,85,45,30)
)
ggplot(climate,
       aes(x=Month,
           y=Temperature,
           group=1))+
  geom_line(color="blue",linewidth=1.2)+
  geom_point(size=3,color="red")+
  labs(title="Monthly Temperature Variation",
       x="Month",
       y="Temperature (°C)")+
  theme_minimal()
```



CODE:

```
ggplot(climate,
       aes(x=Month,
           y=Temperature,
           group=1))+
  geom_line(color="blue",linewidth=1.2)+
  geom_point(size=3,color="red")+
  labs(title="Monthly Temperature Variation",
       x="Month",
       y="Temperature (°C)")+
  theme_minimal()
ggplot(climate,
       aes(x=Month,
           y=Rainfall,
           fill=Rainfall))+
  geom_bar(stat="identity")+
  coord_polar()+
  scale_fill_gradient(low="lightblue",
                      high="darkblue")+
  labs(title="Annual Rainfall Distribution")+
```

theme_minimal()

Comparison

Cartesian Chart

Advantages

- Easy to compare monthly temperature.
- Shows trends clearly.
- Suitable for continuous data.

Limitations

- Does not emphasize yearly cycle.

Polar Chart

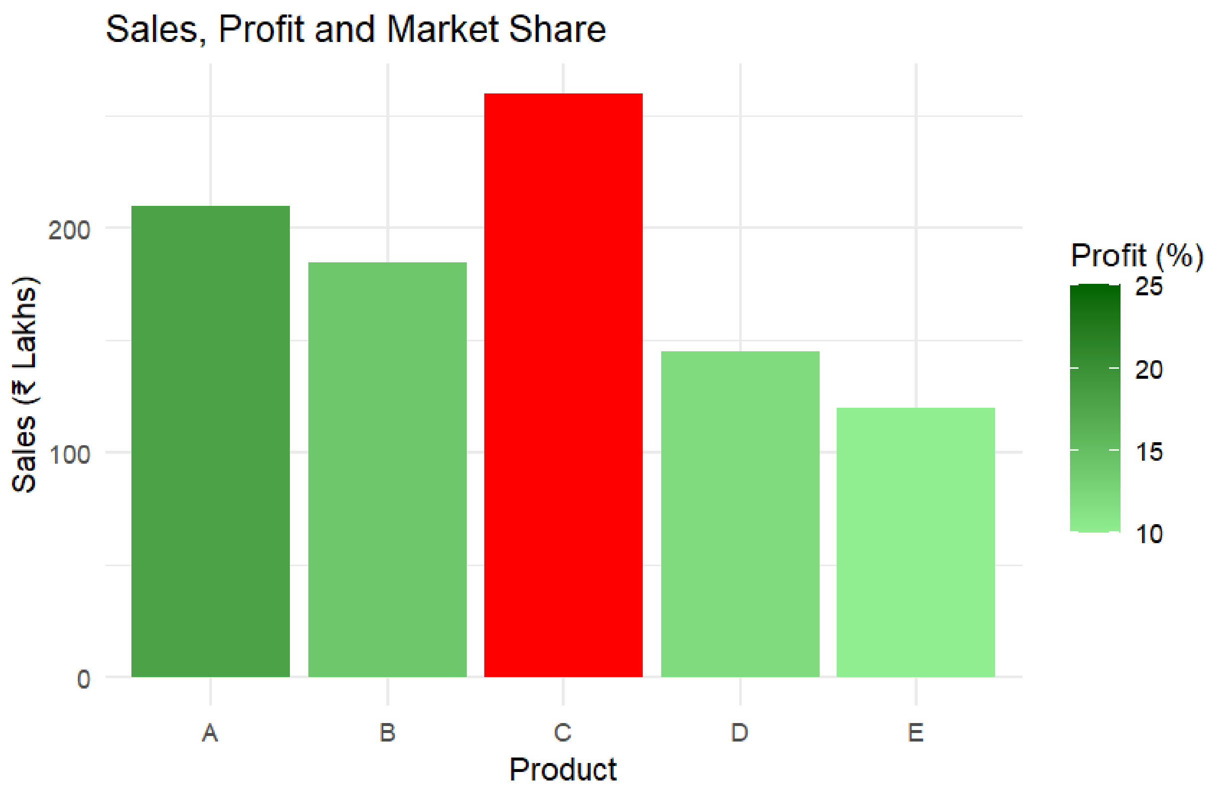
Advantages

- Represents seasonal rainfall patterns.
- Circular layout matches annual climate cycle.
- Visually attractive.

Limitations

- Exact values are harder to compare.
- Less accurate for quantitative comparison.

Question 3 – Business Analytics & Marketing



CODE:

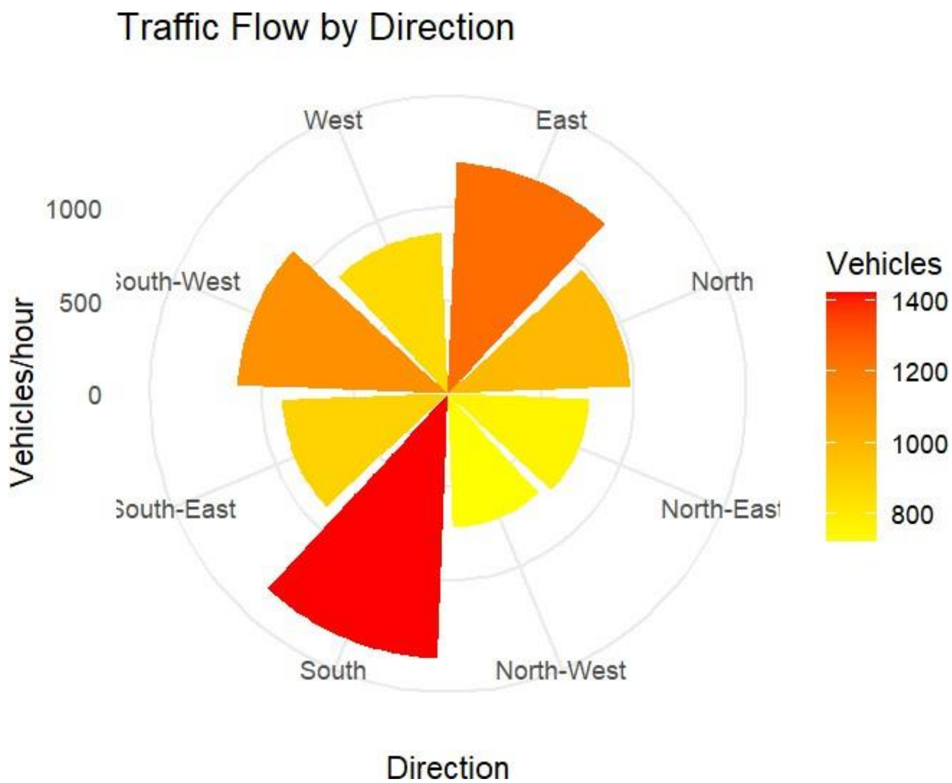
```
library(ggplot2)

business <- data.frame(
  Product=c("A","B","C","D","E"),
  Sales=c(210,185,260,145,120),
  Profit=c(18,14,25,12,10),
  MarketShare=c(24,20,28,15,13)
)

business$Highlight <- ifelse(business$MarketShare==
                              max(business$MarketShare),
                            "Highest","Others")

ggplot(business,
       aes(x=Product,
           y=Sales,
           fill=Profit))+
```

```
geom_bar(stat="identity")+
geom_bar(data=subset(business,Highlight=="Highest"),
         fill="red",
         stat="identity")+
scale_fill_gradient(low="lightgreen",
                    high="darkgreen")+
labs(title="Sales, Profit and Market Share",
     x="Product",
     y="Sales (₹ Lakhs)",
     fill="Profit (%)")+
theme_minimal()
```

Justification

- **Bar Height** represents sales volume.
- **Sequential Green Palette** indicates increasing profit.
- **Red Highlight** immediately identifies the product with the highest market share.
- Helps managers quickly identify the best-performing product.

Question 4 – Transportation Engineering



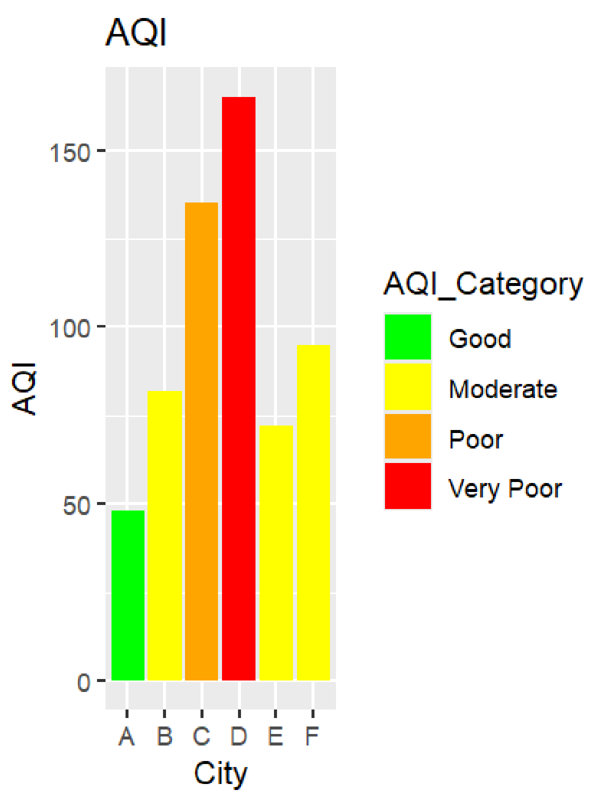
CODE:

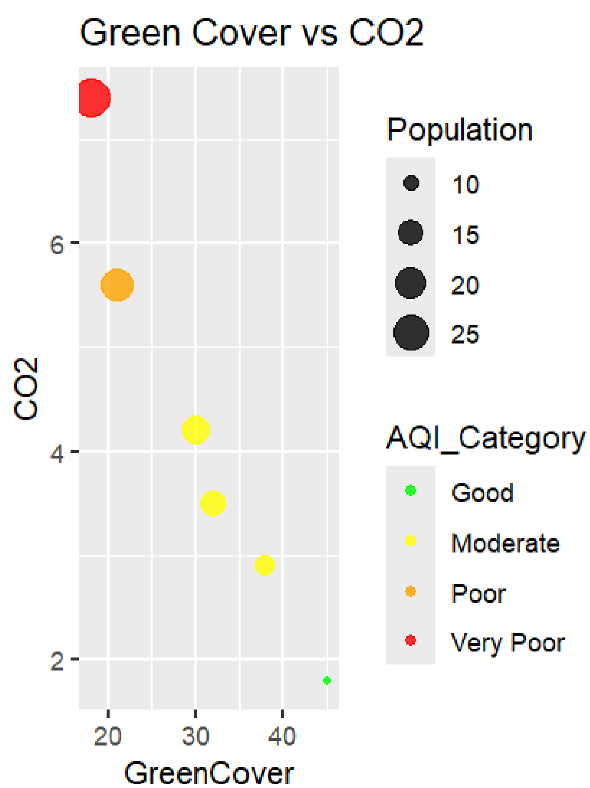
```
library(ggplot2)
traffic <- data.frame(
  Direction=c("North","North-East","East","South-East",
            "South","South-West","West","North-West"),
  Vehicles=c(980,760,1240,890,1420,1130,860,720)
)
ggplot(traffic,
       aes(x=Direction,
           y=Vehicles,
           fill=Vehicles))+
  geom_bar(stat="identity")+
  coord_polar()+
  scale_fill_gradient(low="yellow",
                      high="red")+
  labs(title="Traffic Flow by Direction",
       x="Direction",
       y="Vehicles/hour")+
  theme_minimal()
```

Justification

- Polar chart naturally represents compass directions.
- High-density traffic appears with darker colors.
- Easier to identify heavily congested directions.
- Better than Cartesian charts for directional data.

Question 5 – Environmental Sustainability Dashboard





CODE:

```
library(ggplot2)
library(gridExtra)
env <- data.frame(
  City=c("A","B","C","D","E","F"),
  AQI=c(48,82,135,165,72,95),
  GreenCover=c(45,32,21,18,38,30),
  CO2=c(1.8,3.5,5.6,7.4,2.9,4.2),
  Population=c(9,15,22,28,12,18)
)
```

```r
env$AQI_Category <- cut(
  env$AQI,
  breaks=c(0,50,100,150,200),
  labels=c("Good","Moderate","Poor","Very Poor")
)
# Plot 1
p1 <- ggplot(env,
             aes(x=City,
                 y=AQI,
                 fill=AQI_Category))+
  geom_bar(stat="identity")+
  scale_fill_manual(values=c("green",
                             "yellow",
                             "orange",
                             "red"))+
  labs(title="AQI")
# Plot 2
p2 <- ggplot(env,
             aes(x=GreenCover,
                 y=CO2,
                 size=Population,
                 color=AQI_Category))+
  geom_point(alpha=0.8)+
  scale_color_manual(values=c("green",
                              "yellow",
                              "orange",
                              "red"))+
labs(title="Green Cover vs CO2")
grid.arrange(p1,p2,ncol=2)
```

Justification

- **Bar Chart:** Compares AQI across cities.
- **Scatter Plot:** **S**hows relationship between Green Cover and CO_2 emissions.
- **Point Size:** Represents population.
- **Color Palette:**
    - **Green → Good AQI**
    - **Yellow → Moderate**
    - **Orange → Poor**
    - **Red → Very Poor**
- The dashboard combines multiple environmental indicators into a single view, making it easier to identify pollution hotspots and support sustainability planning.